Day 17

System Verilog: Functional Coverage

Cover points for FIFO:

```
module fifo_with_assertions #(

   parameter DEPTH = 8,
   parameter WIDTH = 8
)(
   input  logic           clk, rst,
   input  logic           wr_en, rd_en,
   input  logic [WIDTH-1:0] din,
   output logic [WIDTH-1:0] dout,
   output logic           full, empty
);
   reg [WIDTH-1:0] mem [0:DEPTH-1];
reg [$clog2(DEPTH):0] wr_ptr, rd_ptr, count;
// write
always @(posedge clk) begin
  if(rst) wr_ptr <= 0;
  else if(wr_en && !full) begin
    mem[wr_ptr] <= din;
    wr_ptr <= (wr_ptr == DEPTH-1) ? 0 : wr_ptr + 1;
  end
 end
 //read
 always @(posedge clk) begin
   if(rst) rd_ptr <=0;
   else if(rd_en && !empty) begin
     dout <= mem[rd_ptr];
     rd_ptr <= (rd_ptr == DEPTH-1) ? 0 : rd_ptr + 1;
  end
 end
 always @(posedge clk) begin
   if(rst) count <= 0;
   else begin
     case({wr_en &!full, rd_en & !empty})
       2'b10: count<= count+1;
       2'b01: count<= count-1;
       2'b11: count<= count;
       default: count<= count;
      endcase
    end
  end

  assign full = (count == DEPTH);
  assign empty = (count == 0);

    // — Assertion 1: never write when full ——————————
    property p_no_write_when_full;
        @(posedge clk) disable iff(rst)
        full |-> !wr_en;
    endproperty
    assert property(p_no_write_when_full)
    else $error("Write attempted when FIFO full");
```

```
  // — Assertion 2: never read when empty ————————————
  property p_no_read_when_empty;
    @(posedge clk) disable iff(rst)
    empty |-> !rd_en;
  endproperty
  assert property(p_no_read_when_empty)
  else $error("Read attempted when FIFO empty");

  // — Assertion 3: full and empty never both true ————————
  property p_not_full_and_empty;
    @(posedge clk) disable iff(rst)
    !(full && empty);
  endproperty
  assert property(p_not_full_and_empty)
   $info("Success: FIFO is not full and empty simulataneously");
   else $error("Fail: FIFO full AND empty simultaneously ie impossible");

  // — Assertion 4: after reset both flags correct ————————
  property p_reset_state;
    @(posedge clk)
    $rose(rst) |=> (empty == 1 && full == 0);
  endproperty
  assert property(p_reset_state)
   $info("Success: Reset");
   else $error("Fail: Wrong state after reset");

  // — Assertion 5: full deasserts after a read ——————————
  property p_full_clears_on_read;
    @(posedge clk) disable iff(rst)
    (full && rd_en) |=> !full;
  endproperty
  assert property(p_full_clears_on_read)
   $info("Success: Full flag cleared the read");
   else $error("Fail: Full flag did not clear after read");

  // — Assertion 6: empty deasserts after a write ————————
  property p_empty_clears_on_write;
    @(posedge clk) disable iff(rst)
    (empty && wr_en) |=> !empty;
  endproperty
  assert property(p_empty_clears_on_write)
   $info("Success: Empty flag cleared after write");
   else $error("Fail: Empty flag did not clear after write");

endmodule
```

Testbench with coverage points:

```
module tb_fifo_coverage;
  parameter DEPTH = 8;
  parameter WIDTH = 8;

  logic          clk, rst;
  logic          wr_en, rd_en;
  logic [WIDTH-1:0] din, dout;
  logic          full, empty;
```

```
  logic [$clog2(DEPTH):0] count; // expose count for coverage

  fifo_with_assertions #(DEPTH, WIDTH) dut(
    .clk(clk), .rst(rst),
    .wr_en(wr_en), .rd_en(rd_en),
    .din(din), .dout(dout),
    .full(full), .empty(empty)
  );

  initial begin clk = 0; forever #5 clk = ~clk; end

  // — Define the covergroup ——————————————————————
  covergroup fifo_cg @(posedge clk);

    // Coverpoint 1: FIFO occupancy levels
    // Did we test empty, quarter-full, half-full, and full?
    cp_count: coverpoint dut.count {
      bins empty       = {0};
      bins quarter_full = {[1:2]};
      bins half_full    = {[3:5]};
      bins nearly_full  = {[6:7]};
      bins full        = {8};
    }

    // Coverpoint 2: write enable states
    cp_wr: coverpoint wr_en {
      bins wr_active   = {1};
      bins wr_inactive = {0};
    }

    // Coverpoint 3: read enable states
    cp_rd: coverpoint rd_en {
      bins rd_active   = {1};
      bins rd_inactive = {0};
    }

    // Coverpoint 4: flag states
    cp_flags: coverpoint {full, empty} {
      bins both_zero  = {2'b00};  // normal operation
      bins empty_only = {2'b01};  // FIFO empty
      bins full_only  = {2'b10};  // FIFO full
      // 2'b11 is illegal — full AND empty simultaneously
      illegal_bins both_set = {2'b11};
    }

    // Coverpoint 5: data values — corner cases
    cp_data: coverpoint din {
      bins zeros    = {8'h00};
      bins ones     = {8'hFF};
      bins walking1 = {8'h01, 8'h02, 8'h04, 8'h08,
                       8'h10, 8'h20, 8'h40, 8'h80};
      bins others   = default;
    }

    // Cross coverage — combinations of coverpoints
    // Did we test writing when full? Reading when empty?
cross_wr_flags: cross cp_wr, cp_flags, cp_rd { // Added cp_rd here so you can use it below
```

```
  // binsof(CoverpointName) intersect {ValueOrBinName}
  bins write_when_full = binsof(cp_wr) intersect {1} && binsof(cp_flags.full_only);

  bins read_when_empty = binsof(cp_rd) intersect {1} && binsof(cp_flags.empty_only);
}

  endgroup

  // Instantiate the covergroup
  fifo_cg cg = new();

  // ── Transaction class ──────────────────────────────
  class fifo_txn;
    rand logic [7:0] data;
    rand logic       wr_en;
    rand logic       rd_en;

    constraint rw_excl  { !(wr_en && rd_en); }
    constraint not_idle { wr_en || rd_en;    }
  endclass

  // ── Main test ──────────────────────────────────────
  initial begin
    automatic fifo_txn txn = new();

   cg.start();

    rst = 1; wr_en = 0; rd_en = 0; din = 0;
    #20; rst = 0;

    // Run random transactions
    repeat(200) begin
      assert(txn.randomize());
      @(negedge clk);
      wr_en = txn.wr_en;
      rd_en = txn.rd_en;
      din   = txn.data;
      @(posedge clk); #1;
      wr_en = 0; rd_en = 0;
    end
    cg.stop();
    // Print coverage report
    $display("\n=== Coverage Report ===");
    $display("Overall coverage: %0.2f%%", cg.get_coverage());
    $display("cp_count  coverage: %0.2f%%", cg.cp_count.get_coverage());
    $display("cp_flags  coverage: %0.2f%%", cg.cp_flags.get_coverage());
    $display("cp_data   coverage: %0.2f%%", cg.cp_data.get_coverage());

    #20; $finish;
  end
endmodule
```

Output:

```
xrun: 25.03-s001: (c) Copyright 1995-2025 Cadence Design Systems, Inc.
        Top level design units:
```

tb_fifo_coverage

```
xcelium> source /xcelium25.03/tools/xcelium/files/xmsimrc
xcelium> run
xmsim: *N,ASRTST (./design.sv,76): (time 15 NS) Assertion tb_fifo_coverage.dut.p_reset_state has passed
Success: Reset
xmsim: *N,ASRTST (./design.sv,67): (time 25 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,94): (time 35 NS) Assertion tb_fifo_coverage.dut.p_empty_clears_on_write has
passed
Success: Empty flag cleared after write
xmsim: *N,ASRTST (./design.sv,67): (time 35 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 45 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,94): (time 55 NS) Assertion tb_fifo_coverage.dut.p_empty_clears_on_write has
passed
Success: Empty flag cleared after write
xmsim: *N,ASRTST (./design.sv,67): (time 55 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 65 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 75 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 85 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,59): (time 85 NS) Assertion tb_fifo_coverage.dut.p_no_read_when_empty has failed
Read attempted when FIFO empty
xmsim: *N,ASRTST (./design.sv,67): (time 95 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,94): (time 105 NS) Assertion tb_fifo_coverage.dut.p_empty_clears_on_write has
passed
Success: Empty flag cleared after write
xmsim: *N,ASRTST (./design.sv,67): (time 105 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 115 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 125 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 135 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 145 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 155 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 165 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 175 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 185 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 195 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 205 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 215 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
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xmsim: *N,ASRTST (./design.sv,67): (time 225 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 235 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 245 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 255 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 265 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 275 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 285 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 295 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,59): (time 295 NS) Assertion tb_fifo_coverage.dut.p_no_read_when_empty has failed
Read attempted when FIFO empty
xmsim: *N,ASRTST (./design.sv,67): (time 305 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,59): (time 305 NS) Assertion tb_fifo_coverage.dut.p_no_read_when_empty has failed
Read attempted when FIFO empty
xmsim: *N,ASRTST (./design.sv,67): (time 315 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,59): (time 315 NS) Assertion tb_fifo_coverage.dut.p_no_read_when_empty has failed
Read attempted when FIFO empty
xmsim: *N,ASRTST (./design.sv,67): (time 325 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,59): (time 325 NS) Assertion tb_fifo_coverage.dut.p_no_read_when_empty has failed
Read attempted when FIFO empty
xmsim: *N,ASRTST (./design.sv,67): (time 335 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,94): (time 345 NS) Assertion tb_fifo_coverage.dut.p_empty_clears_on_write has
passed
Success: Empty flag cleared after write
xmsim: *N,ASRTST (./design.sv,67): (time 345 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 355 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 365 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 375 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 385 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 395 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 405 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 415 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 425 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 435 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 445 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
```

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xmsim: *N,ASRTST (./design.sv,67): (time 455 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 465 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 475 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 485 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 495 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,59): (time 495 NS) Assertion tb_fifo_coverage.dut.p_no_read_when_empty has failed
Read attempted when FIFO empty
xmsim: *N,ASRTST (./design.sv,67): (time 505 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,94): (time 515 NS) Assertion tb_fifo_coverage.dut.p_empty_clears_on_write has
passed
Success: Empty flag cleared after write
xmsim: *N,ASRTST (./design.sv,67): (time 515 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 525 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 535 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 545 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 555 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 565 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 575 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 585 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 595 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 605 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 615 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 625 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 635 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 645 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 655 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 665 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 675 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 685 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 695 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 705 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
```

```
xmsim: *N,ASRTST (./design.sv,67): (time 715 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 725 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 735 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 745 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 755 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 765 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 775 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 785 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 795 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 805 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 815 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 825 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 825 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 835 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 835 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 845 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 845 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 855 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 855 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 865 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 865 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 875 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 875 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 885 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 885 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 895 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 895 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 905 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 915 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
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Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 915 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 925 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 935 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 935 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 945 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 955 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 965 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 975 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 985 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 995 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1005 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1005 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1015 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1015 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1025 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1035 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1035 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1045 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1055 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1055 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1065 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1075 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1085 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1095 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has

passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1105 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1115 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1115 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1125 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1135 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1135 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1145 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1155 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1155 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1165 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1175 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1175 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1185 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1195 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1195 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1205 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1215 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1225 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1235 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1235 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1245 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has

passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1255 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1255 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1265 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1275 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1285 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1295 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1305 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1315 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1325 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1325 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1335 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1345 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1345 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1355 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1365 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1375 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1385 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1385 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1395 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1405 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read

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xmsim: *N,ASRTST (./design.sv,67): (time 1405 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1415 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1425 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1435 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1435 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1445 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1445 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1455 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1455 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1465 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1465 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1475 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1485 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1485 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1495 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1495 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1505 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1515 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1515 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1525 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1525 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1535 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
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Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1535 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1545 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1545 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1555 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1565 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1565 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1575 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1585 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1595 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1605 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1605 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1615 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1615 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1625 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1635 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1635 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1645 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1655 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1665 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1675 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1685 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed

Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1695 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1705 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1705 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1715 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1725 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1725 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1735 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1745 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1755 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1765 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1775 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1785 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1795 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1805 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1815 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1825 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1835 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1845 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1855 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1865 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has

```
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1875 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1885 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1885 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1895 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *E,ASRTST (./design.sv,51): (time 1895 NS) Assertion tb_fifo_coverage.dut.p_no_write_when_full has failed
Write attempted when FIFO full
xmsim: *N,ASRTST (./design.sv,67): (time 1905 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,85): (time 1915 NS) Assertion tb_fifo_coverage.dut.p_full_clears_on_read has passed
Success: Full flag cleared the read
xmsim: *N,ASRTST (./design.sv,67): (time 1915 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1925 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1935 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1945 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1955 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1965 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1975 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1985 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 1995 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 2005 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 2015 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has
passed
Success: FIFO is not full and empty simulataneously

=== Coverage Report ===
Overall coverage: 89.44%
cp_count  coverage: 100.00%
```

```
cp_flags  coverage: 100.00%
cp_data   coverage: 66.67%
xmsim: *N,ASRTST (./design.sv,67): (time 2025 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
xmsim: *N,ASRTST (./design.sv,67): (time 2035 NS) Assertion tb_fifo_coverage.dut.p_not_full_and_empty has passed
Success: FIFO is not full and empty simulataneously
Simulation complete via $finish(1) at time 2036 NS + 0
```

The coverage report shows the overall coverage here which is 89 % , only the cover point of data needs to be covered.

To increase the coverage percentage, I am explicitly adding few more testcases here

```
repeat(200) begin
        assert(txn.randomize());
        @(negedge clk);
        wr_en = txn.wr_en & !full;
                        rd_en = txn.rd_en & !empty;
        din   = txn.data;
        @(posedge clk); #1;
        wr_en = 0; rd_en = 0;
        wr_en = 1; din = 8'h00;  // force write zero
                        @(negedge clk); wr_en = 0;
        @(negedge clk);
                        wr_en = 1; din = 8'hFF;
                        @(negedge clk); wr_en = 0;
    end
```

Output:

```
xcelium> run

=== Coverage Report ===
Overall coverage: 96.67%
cp_count  coverage: 100.00%
cp_flags  coverage: 100.00%
cp_data   coverage: 100.00%
```

We can see that the coverage has increased to 96.67%

Coverage-driven loop: running until covered, here the coverage will be taking place till it reaches 95% coverage point or 10,000 iterations

```
module tb_fifo_coverage;
   parameter DEPTH = 8;
   parameter WIDTH = 8;

   logic clk, rst;
   logic wr_en, rd_en;
   logic [WIDTH-1:0] din, dout;
   logic full, empty;

   // Instance of the FIFO
   fifo #(DEPTH, WIDTH) dut (
      .clk(clk),
```

```
    .rst(rst),
    .wr_en(wr_en),
    .rd_en(rd_en),
    .din(din),
    .dout(dout),
    .full(full),
    .empty(empty)
);

// Clock Generation
initial begin
    clk = 0;
    forever #5 clk = ~clk;
end

// — Define the covergroup ——————————————————
covergroup fifo_cg @(posedge clk);
    // Coverpoint 1: FIFO occupancy levels
    cp_count: coverpoint dut.count {
        bins empty = {0};
        bins quarter_full = {[1:2]};
        bins half_full = {[3:5]};
        bins nearly_full = {[6:7]};
        bins full = {8};
    }

    // Coverpoint 2: write enable states
    cp_wr: coverpoint wr_en {
        bins wr_active = {1};
        bins wr_inactive = {0};
    }

    // Coverpoint 3: read enable states
    cp_rd: coverpoint rd_en {
        bins rd_active = {1};
        bins rd_inactive = {0};
    }

    // Coverpoint 4: flag states
    cp_flags: coverpoint {full, empty} {
        bins both_zero = {2'b00};
        bins empty_only = {2'b01};
        bins full_only = {2'b10};
        illegal_bins both_set = {2'b11};
    }

    // Coverpoint 5: data values
    cp_data: coverpoint din {
        bins zeros = {8'h00};
        bins ones = {8'hFF};
        bins walking1 = {8'h01, 8'h02, 8'h04, 8'h08, 8'h10, 8'h20, 8'h40, 8'h80};
        bins others = default;
    }

    // Cross coverage
    cross_wr_flags: cross cp_wr, cp_flags, cp_rd {
        bins write_when_full = binsof(cp_wr) intersect {1} && binsof(cp_flags.full_only);
```

```
    bins read_when_empty = binsof(cp_rd) intersect {1} && binsof(cp_flags.empty_only);
  }
endgroup

// Instantiate the covergroup
fifo_cg cg = new();

// — Transaction class ————————————————————
class fifo_txn;
  rand logic [7:0] data;
  rand logic wr_en;
  rand logic rd_en;
  constraint rw_excl { !(wr_en && rd_en); }
  constraint not_idle { wr_en || rd_en; }
endclass

// — Main Simulation Block ————————————————————
initial begin
  // Variables MUST be declared at the start of the block
  automatic fifo_txn txn = new();
  int unsigned iterations = 0;
  real coverage = 0.0;

  // Initialize signals
  rst = 1;
  wr_en = 0;
  rd_en = 0;
  din = 0;

  #20;
  rst = 0;
  cg.start();

  $display("Starting simulation...");

  // Run until 95% coverage or 10,000 iterations
 while(coverage < 95.0 && iterations < 10000) begin
    if(!txn.randomize()) $fatal("Randomization failed");

    @(negedge clk);
    // Drive signals (with simple overflow/underflow protection)
    wr_en = txn.wr_en;
    rd_en = txn.rd_en;
    din = txn.data;

    @(posedge clk);
    #1; // Wait for sampling to occur

    coverage = cg.get_coverage();
    iterations = iterations + 1;

    if(iterations % 500 == 0)
      $display("Iteration %0d: Current Coverage = %0.2f%%", iterations, coverage);
  end

  cg.stop();
```

```
    // — Final Coverage Report ————————————————————
    $display("\n========================================");
    $display("   FINAL COVERAGE REPORT");
    $display("========================================");
    $display("Overall coverage:  %0.2f%%", cg.get_coverage());
    $display("Occupancy (count): %0.2f%%", cg.cp_count.get_coverage());
    $display("Flags coverage:    %0.2f%%", cg.cp_flags.get_coverage());
    $display("Data coverage:     %0.2f%%", cg.cp_data.get_coverage());
    $display("Total Iterations:  %0d", iterations);
    $display("========================================\n");

   if(coverage >= 95.0)
     $display("RESULT: PASS: Coverage goal reached!");
    else
     $display("RESULT: FAIL: Goal not met within iteration limit.");

    #20;
    $finish;
  end

endmodule
```

Output:

```
xcelium> run
Starting simulation...

========================================
  FINAL COVERAGE REPORT
========================================
Overall coverage:  95.00%
Occupancy (count): 100.00%
Flags coverage:    100.00%
Data coverage:     100.00%
Total Iterations:  57
========================================

RESULT: PASS: Coverage goal reached!
```